MACHINE LEARNING MINI PROJECT

Road Accident Severity Prediction Using Machine Learning

Comparative Classification and Class-Imbalance Analysis

Project Report

Submitted by: Anant Santosh Yadav.
MIS No. : 712652074
Branch : MTech Data Science
Institute: COEP Technological University

Guide : Dr. Y. V. Haribhakta.

Abstract

This project develops a supervised machine-learning workflow to classify the recorded severity of a road accident into Fatal injury, Serious Injury, or Slight Injury. The dataset was loaded from Road.csv. Twenty-one accident, driver, vehicle, road, lighting, and weather attributes were selected as input features, while Accident_severity was used as the target. Text placeholders were standardized, rows without a target label were removed, numeric and categorical variables were handled through a preprocessing pipeline, and the data was split using a stratified 80:20 train/test split. Six classifiers were compared using five-fold stratified cross-validation. Additional experiments evaluated SVM hyperparameter selection and class-imbalance handling through RandomOverSampler and Balanced Random Forest. The final class-weighted Logistic Regression achieved 66.84% accuracy, 63.78% balanced accuracy, and a Macro F1-score of 0.4526 on the held-out test set. Its recall was 68% for Fatal injury and 55% for Serious Injury. These results show why accuracy alone is insufficient for imbalanced accident-severity data. This is a classifier for already-recorded accidents, not a predictor of whether an accident will occur or a substitute for professional road-safety decisions.

1. Introduction and Problem Statement

Road accidents vary in severity, from slight injuries to serious injuries and fatalities. A data-driven classifier can help analyze how recorded accident circumstances relate to the severity label. However, accident-severity datasets are often imbalanced: slight-injury records can greatly outnumber fatal-injury records. A model can therefore obtain apparently strong accuracy while failing to identify severe cases.

This project builds and evaluates a multi-class classification pipeline using driver, vehicle, time, road, lighting, and weather information. It classifies the severity recorded for an accident; it does not predict whether an accident will happen. Number_of_vehicles_involved is generally known only after an accident has occurred, so the model is not a purely pre-accident warning system.

2. Objectives

- Load and inspect Road.csv and the Accident_severity target.
- Clean common text placeholders and remove records without target labels.
- Prepare 21 input features and build a reusable preprocessing pipeline.
- Compare classifiers using five-fold stratified cross-validation.
- Tune SVM hyperparameters and compare class-imbalance experiments.
- Evaluate the selected model using accuracy, balanced accuracy, Macro F1, per-class metrics, and a confusion matrix.
- Save the trained model and experiment tables.

3. Literature Review

Prior research on crash-severity modelling identifies class imbalance as a recurring challenge, especially when fatal or otherwise severe outcomes are rare. Fiorentini and Losa (2020) examine methods for handling imbalanced road-crash severity data. Chen et al. (2025) study prediction and interpretation of crash severity with imbalanced traffic-crash data. Amiri, Afshari, and Soltani (2025) compare machine-learning and imbalance-handling methods for traffic-accident severity prediction, with particular attention to rare fatal outcomes.

This project follows that practical direction by comparing standard classifiers, using class weighting where supported, and evaluating resampling-based experiments. It does not claim a new classification algorithm; its

focus is a reproducible model-comparison workflow and clear evaluation of minority-class recall alongside accuracy.

4. Dataset and Features

The notebook loads a CSV file named Road.csv and uses Accident_severity as the target label. The target classes are Fatal injury, Serious Injury, and Slight Injury. The test split contains 2,464 observations, corresponding to 20% of the cleaned labelled records; the training set contains 9,856 observations.

Feature group	Selected fields / examples	Purpose
Time and driver	Time, Day_of_week, Age_band_of_driver, Sex_of_driver, Educational_level, Driving_experience	Represent timing and driver characteristics
Vehicle	Vehicle_driver_relation, Type_of_vehicle, Owner_of_vehicle, Service_year_of_vehicle, Defect_of_vehicle, Number_of_vehicles_involved, Vehicle_movement	Represent vehicle context
Road and environment	Area_accident_occured, Lanes_or_Medians, Road_allignment, Types_of_Junction, Road_surface_type, Road_surface_conditions, Light_conditions, Weather_conditions	Represent road and environmental context

The selected input list contains 21 features. Casualty-related fields and Cause_of_accident are excluded by the notebook's feature-selection design. Number_of_vehicles_involved is converted to numeric; invalid values become missing. Common text placeholders such as “na”, “n/a”, “unknown”, “none”, and “?” are standardized as missing values. Rows without Accident_severity are removed, while missing feature values are imputed in the preprocessing pipeline.

5. System Architecture and Methodology

5.1 Data preprocessing

Numeric features are imputed with the median. Categorical features are imputed with the most frequent value and one-hot encoded. Preprocessing is included inside each model pipeline so it is fitted on the relevant training fold during cross-validation.

5.2 Train/test split and validation

A stratified 80:20 split is created with random_state=42: 9,856 training records and 2,464 test records. Five-fold StratifiedKFold cross-validation is used on the training set for model comparison and SVM tuning. Macro F1 and balanced accuracy are considered alongside accuracy because the target classes are imbalanced.

Model	Configuration / role
Majority Baseline	Predicts the most frequent training class
Decision Tree	Class-weighted tree, max_depth=10, min_samples_leaf=10
Random Forest	150 trees; class_weight='balanced'
Gradient Boosting	GradientBoostingClassifier baseline
Logistic Regression	class_weight='balanced'; max_iter=2000
SVM	SVC with class_weight='balanced'; C values 0.1, 1, and 10 evaluated

5.3 Class-imbalance experiments

Two additional approaches were evaluated on the same train/test split: Random Forest with RandomOverSampler applied within the training pipeline, and Balanced Random Forest. Resampling was applied only to training data. A separate class-weighted Logistic Regression experiment was then fitted and saved as the final model because it achieved the strongest balanced accuracy and fatal/serious injury recall among the reported held-out test candidates.

6. Results and Discussion

6.1 Five-fold cross-validation model comparison

Model	Accuracy	Balanced accuracy	Macro F1
Gradient Boosting	0.8507	0.4350	0.4510
SVM	0.6941	0.5003	0.4808
Logistic Regression	0.6680	0.6162	0.4487
Random Forest	0.8458	0.3340	0.3069
Decision Tree	0.6808	0.4736	0.3780
Majority Baseline	0.8456	0.3333	0.3055

Gradient Boosting recorded the highest mean accuracy in cross-validation, but its balanced accuracy was lower than Logistic Regression's. The majority baseline also achieved high accuracy because Slight Injury dominates the dataset; its balanced accuracy was approximately one-third. This shows why model selection should not rely on accuracy alone.

6.2 SVM hyperparameter tuning

SVM C	Accuracy	Balanced accuracy	Macro F1
10	0.8101	0.4913	0.5220
1	0.6941	0.5003	0.4808
0.1	0.5570	0.4666	0.3410

Among tested SVM settings, C=10 achieved the highest Macro F1 (0.5220) and was the best SVM configuration by that tuning criterion. The SVM tuning table is retained as experimental evidence, although the final saved model was later changed to class-weighted Logistic Regression.

6.3 Final balanced Logistic Regression results

Metric	Test result
Accuracy	0.6684 (66.84%)
Balanced accuracy	0.6378 (63.78%)
Macro F1	0.4526
Fatal injury recall	0.68
Serious Injury recall	0.55
Slight Injury recall	0.69

Class	Precision	Recall	F1-score	Support
Fatal injury	0.13	0.68	0.22	31
Serious Injury	0.27	0.55	0.36	349
Slight Injury	0.91	0.69	0.78	2,084
Macro average	0.43	0.64	0.45	2,464
Weighted average	0.81	0.67	0.72	2,464

The final class-weighted Logistic Regression achieved 66.84% accuracy, 63.78% balanced accuracy, and 0.4526 Macro F1 on the test set. It detected 68% of Fatal injury cases and 55% of Serious Injury cases, while recall for Slight Injury was 69%. Fatal-injury precision is low (0.13), meaning many predictions of Fatal injury are false positives. This limitation matters even though fatal-injury recall is higher.

7. Class-Imbalance Experiment Results

Experiment	Accuracy	Balanced accuracy	Macro F1
Balanced Logistic Regression (test)	0.6684	0.6378	0.4526
Random Forest + RandomOverSampler (test)	0.8498	0.3731	0.3799
Balanced Random Forest (test)	0.6485	0.5070	0.4060

These are held-out test results. The SVM tuning table reports cross-validation scores, so those values should not be mixed into the test-set leaderboard. Random Forest + RandomOverSampler recorded 84.98% accuracy but only 10% fatal-injury recall and 2% serious-injury recall. Balanced Random Forest achieved 42% fatal-injury recall and 41% serious-injury recall, with lower accuracy. Balanced Logistic Regression achieved the highest test balanced accuracy among these reported candidates.

8. Evaluation Metrics

Metric	Interpretation
Accuracy	Fraction of all predictions that are correct; can be dominated by the majority class.
Balanced accuracy	Average recall across classes; useful when class sizes differ.
Macro F1	Unweighted mean of class-wise F1 scores; gives each class equal weight.

Precision	Of predictions for a class, the fraction that are correct.
Recall	Of actual records in a class, the fraction detected.
Confusion matrix	Shows which severity classes are confused with each other.

Macro F1 and balanced accuracy are especially relevant because fatal-injury cases form a small part of the test set. Per-class precision and recall reveal limitations that a single accuracy figure can hide.

9. Limitations

- The target classes are strongly imbalanced, and Fatal injury has only 31 test examples; class-wise estimates may vary with another split.
- Fatal injury precision is 0.13, so the model produces many false positive Fatal injury predictions.
- The model classifies recorded accidents; it does not predict whether an accident will occur.
- Number_of_vehicles_involved is generally known after an accident, so this is not a purely pre-accident warning system.
- Results are based on one dataset and one split; external validation on independently collected data is needed.
- Cross-validation and test metrics come from different evaluation procedures and must not be conflated.
- The model should not be the sole basis for emergency response, legal, insurance, or road-safety decisions.

10. Conclusion and Future Work

This project implemented a classification workflow for recorded road-accident severity, including data cleaning, feature selection, preprocessing, cross-validation model comparison, SVM tuning, and class-imbalance experiments. The final class-weighted Logistic Regression achieved 66.84% test accuracy, 63.78% balanced accuracy, and a Macro F1-score of 0.4526. Its Fatal injury recall was 68% and Serious Injury recall was 55%, showing improved sensitivity to minority classes at the cost of lower precision and reduced performance on Slight Injury records.

Future work should evaluate repeated stratified splits or nested cross-validation, tune class-specific decision thresholds using validation data only, examine precision–recall curves, test additional cost-sensitive models, and validate on an external dataset. Feature-importance and explainability methods could also be added, with care not to interpret model associations as causal effects.

References

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